(Write-Ups for Questions 3-5)

3. Polynomial norms

For $\mathcal{P}_2$ on $[-1, 1]$ what are $||p||_2$, $||p||_1$, and $||p||_\infty$ for $p(x) = x$?

Using the Polynomial Inner Product, and the Minkowski Definition of Inner Products, we get that for $p(x) = x$, $||p||_2 = \sqrt{\int_{-1}^{1} |x|^2 \, dx} = \sqrt{\int_{-1}^{1} x^2 \, dx} = \sqrt{2/3}$,
$||p||_1 = \int_{-1}^{1} |x| \, dx = 2 * \int_0^1 x \, dx = 1$, $||p||_\infty = \max_{x \in [-1,1]} |x| = 1$

4. Norm constants

For $v \in \mathbb{R}^n$, argue that

$$||v||_\infty \leq ||v||_2 \leq \sqrt{n}||v||_\infty.$$

Note that $||v||_\infty$ is the "Max Norm" and $||v||_2$ is the "Euclidean Norm".

For any vector $v \in \mathbb{R}^n$, we know that one of these elements, let's call it $|v_k|$ is the "max" element, with the largest absolute value.

We know that by definition of the norm-∞: $||v||_\infty = |v_k|$, and per the definition of the Euclidean Norm and the fact that $v_i^2 \geq 0$ for all other $i \neq k$:

$$||v||_2 = \sqrt{\sum_{i=1}^{n} v_i^2} \geq \sqrt{v_k^2} = |v_k| = ||v||_\infty$$

which gives us the first inequality.

For any vector of dimension n, by definition each component $v_i \leq v_k$, thus

$$||v||_2 = \sqrt{\sum_{i=1}^{n} v_i^2} \leq \sqrt{\sum_{i=1}^{n} v_k^2} = \sqrt{n}|v_k| = \sqrt{n}||v||_\infty$$

which gives us the second inequality.

5. Consistency of Frobenius norm

Argue that for any $A \in \mathbb{R}^{m\times n}$ and $x \in \mathbb{R}^n$,

$$\|Ax\|_2 \le \|A\|_F \|x\|_2.$$

By definition, we have

$$\|Ax\|_2^2 = \sum_{i=1}^{m}(\sum_{j=1}^{n} a_{ij}x_j)^2$$

Using Cauchy-Schwarz Inequality (specifically the vectors $u, v \in \mathbb{R}^n$, which has it for the standard inner product that $|u^Tv| \le ||u||_2 \cdot ||v||_2$) (Obviously squaring both sides won't change the inequality)

(Here, let $u = (a_{i1} \ldots a_{in})^T$ which is the $i-th$ row of A as a column vector and $v = x$)

$$(\sum_{j=1}^{n} a_{ij}x_j)^2 = |u^Tv|^2 \le (||u||_2 \cdot ||v||_2)^2 = \sum_{j=1}^{n} a_{ij}^2 \cdot \sum_{j=1}^{n} x_j^2$$

We then have

$$\|Ax\|_2^2 \le \sum_{i=1}^{m}(\sum_{j=1}^{n} a_{ij}^2 \cdot \sum_{j=1}^{n} x_j^2)$$

In $\sum_{i=1}^{m}(\sum_{j=1}^{n} a_{ij}^2 \cdot \sum_{j=1}^{n} x_j^2)$, as $\sum_{j=1}^{n} x_j^2$ is a constant for the outer sum indexed by i, we can thus factor it out of the sum, that is: $\sum_{i=1}^{m}(\sum_{j=1}^{n} \cdot a_{ij}^2 \sum_{j=1}^{n} x_j^2) = (\sum_{i=1}^{m}\sum_{j=1}^{n} a_{ij}^2) \cdot (\sum_{j=1}^{n} x_j^2)$,

therefore, using the defintion of Frobenius norm and Euclidean norm, we have

$$\|Ax\|_2^2 \le (\sum_{i=1}^{m}\sum_{j=1}^{n} a_{ij}^2)(\sum_{j=1}^{n} x_j^2) = \|A\|_F^2 \cdot \|x\|_2^2$$

, we thus proved the inequality.